

Cover Page

Standard: BACB30YP Rev: (W) 31-Aug-2015

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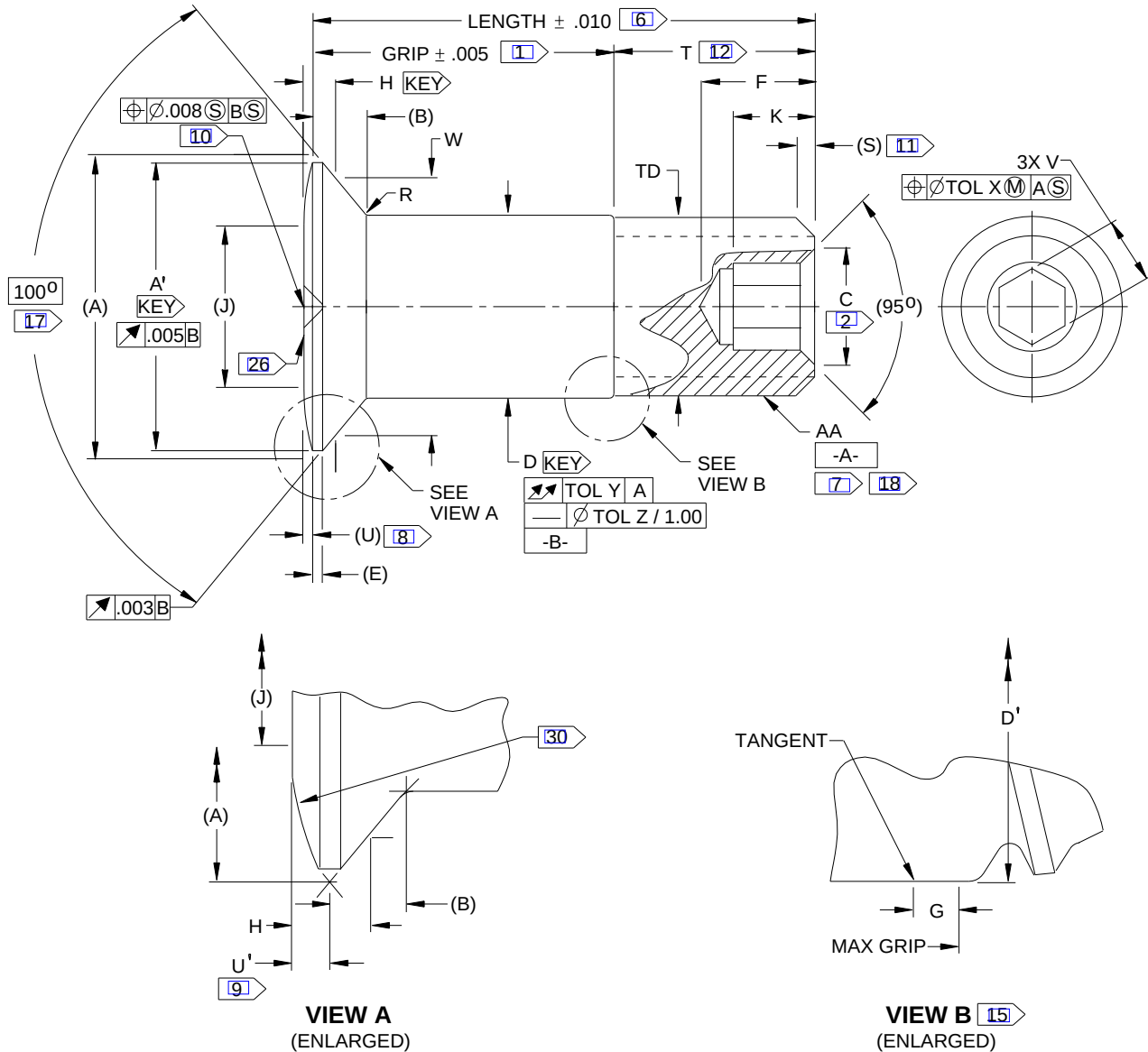
BOEING PART STANDARD
THE BOEING COMPANY

BACB30YP

Revision W
31 August 2015
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BOLT, CLOSE TOLERANCE, 100 DEG REDUCED SHEAR DOME HEAD, HEX DRIVE, 95 KSI SHEAR, 6AL-4V TITANIUM, LIGHTWEIGHT

TECHNICAL CHANGES IDENTIFIED BY REVISION BAR.



DIMENSIONING AND TOLERANCING PER [ANSI Y14.5M-1982](#).

DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.

DIMENSIONS APPLY AFTER FINISH UNLESS OTHERWISE SPECIFIED. [27]

TABLE I - DIMENSIONS

BOEING STANDARD NUMBER BACB30YP 3	AA NOMINAL THREAD SIZE 7	Ø A TO SHARP CORNER REF 4 9		Ø A' HEAD KEY		B HEAD HEIGHT REF 4 9		Ø C	
		LMC	MMC	MAX	MIN	LMC	MMC	MAX	MIN
5	.1640-32	.2532	.2605	.2460	.2410	.0376	.0411	2	2
6	.1900-32	.2937	.3011	.2865	.2815	.0437	.0472	.119	.104
8	.2500-28	.3871	.3944	.3795	.3745	.0577	.0612	.142	.122
10	.3125-24	.4658	.4731	.4550	.4500	.0645	.0680	.180	.160
12	.3750-24	.5519	.5592	.5415	.5365	.0744	.0779	.217	.197

BOEING STANDARD NUMBER BACB30YP ³	Ø D			Ø D' MIN	E LAND REF	F MAX	G MAX	H	
	BEFORE COATING MIN	AFTER COATING						GAGE PROTRUSION	
		KEY						5	
		MAX	MIN					MAX	MIN
5	.1621	.1635	.1625	.1601	.010	.140	.013	.0286	.0256
6	.1881	.1895	.1885	.1857	.010	.140	.016	.0282	.0252
8	.2481	.2495	.2485	.2451	.010	.160	.021	.0316	.0286
10	.3106	.3120	.3110	.3068	.012	.200	.026	.0346	.0316
12	.3731	.3745	.3735	.3693	.012	.235	.030	.0385	.0355

BOEING STANDARD NUMBER BACB30YP 3	K MIN	Ø J FLAT REF		JT RADIUS 14		L MAX (SEE Figure 2) 20	R FILLET RADIUS	
		MAX	MIN	MAX	MIN		MAX	MIN
5	.080	.130	.120	.080	.070	.026	.025	.015
6	.080	.161	.151	.100	.070	.026	.030	.020
8	.090	.195	.185	.140	.130	.027	.030	.020
10	.110	.255	.245	.170	.160	.029	.040	.030
12	.140	.317	.307	.230	.160	.029	.040	.030

BOEING STANDARD NUMBER BACB30YP 3	S REF	T 12	THREADS				U REF 4 8	U' 9	
			Ø TD 7		PITCH DIAMETER KEY				
			MAX	MIN	MAX	MIN		MAX	MIN
5	.031	.280	.1595	.1570	.1437	.1415	.0035	.0053	.0033
6	.031	.290	.1840	.1810	.1697	.1674	.0035	.0053	.0033
8	.031	.320	.2440	.2410	.2268	.2243	.0045	.0062	.0042
10	.047	.380	.3060	.3020	.2854	.2827	.0050	.0069	.0049
12	.047	.420	.3680	.3640	.3479	.3450	.0065	.0085	.0065

TABLE I - DIMENSIONS (Continued)

BOEING STANDARD NUMBER BACB30YP 3	V HEX		Ø W GAGE		TOLERANCES			SHORTEST GRIP NUMBER AVAILABLE
	MAX	MIN	MAX	MIN	X	Y	Z	
5	.0806	.0791	.2028	.2026	.010	.0045	.0040	2
6	.0806	.0791	.2441	.2439	.010	.0045	.0040	2
8	.0967	.0947	.3315	.3313	.010	.0045	.0030	2
10	.1295	.1270	.4047	.4045	.012	.0045	.0030	3
12	.1617	.1582	.4854	.4852	.014	.0060	.0025	4

NOTES

- 1 TO DETERMINE THE NOMINAL GRIP LENGTH, MULTIPLY THE GRIP LENGTH NUMBER BY .0625. THE GRIP LENGTH IS MEASURED FROM THE THEORETICAL INTERSECTION OF THE CROWN RADIUS AND THE 100 DEGREE HEAD ANGLE TO THE END OF THE FULL CYLINDRICAL PORTION OF THE SHANK PLUS THE "G" DIMENSION. GRIP LENGTH, TRANSITION AND RADIUS INSPECTION ("G" AND "D") REQUIRE USE OF AN OVERLAY.
- TO DETERMINE THE GRIP LENGTH NUMBER FOR AN APPLICATION, DIVIDE THE TOTAL THICKNESS OF PARTS BEING JOINED BY .0625 AND ROUND OFF THE DECIMAL TO THE NEXT LARGER WHOLE NUMBER.
- 2 ON SIZE 5 ONLY, COUNTERSINK TO A DEPTH TO BREAK CORNERS OF THE DRIVE HEX.
- 3 SEE CODING SECTION UNDER USAGE AND APPLICATION FOR COMPLETE BOEING PART NUMBER.
- 4 DIMENSIONS "A", "B", "J" AND "U" ARE INCLUDED FOR ENGINEERING REFERENCE ONLY, AND ARE NOT TO BE USED FOR INSPECTION. THEY ARE CALCULATED LIMITS RESULTING FROM THE TOLERANCES ON "D", "W", "H", "E", AND "U".
- 5 HEAD PROTRUSION MEASUREMENT IN ACCORDANCE WITH [D-11805](#), INSPECTION METHOD A.
- 6 APPLY TOLERANCE TO A NOMINAL LENGTH DETERMINED BY ADDING THE NOMINAL GRIP LENGTH (SEE 1) TO "T" FROM [Table I](#).
- 7 THREAD CLASS UNJF-3A EXCEPT UNJC-3A FOR SIZE 5 (.1640-32) PER [AS8879](#). PITCH DIAMETER SHALL BE PER [Table I](#). MAJOR DIAMETER SHALL CONFORM TO "TD".
- 8 MEASURED FROM THE TOP OF THE "E" LAND.
- 9 MEASURED FROM THE THEORETICAL INTERSECTION OF CROWN RADIUS AND 100 DEGREE HEAD ANGLE (SEE VIEW A).
- 10 CENTER DIMPLE IN HEAD IS REQUIRED; DIAMETER .035 MAXIMUM (REF), DEPTH .010 MAXIMUM.
- 11 POINT SHALL BE FLAT AND CHAMFERED. CHAMFER "S" PLUS INCOMPLETE THREADS SHALL NOT EXCEED TWO THREAD PITCHES.

- 12 THE "T" DIMENSION IS A NOMINAL VALUE FOR USE IN ESTABLISHING THE NOMINAL BOLT LENGTH AND IS NOT TO BE SEPARATELY INSPECTED (SEE 6).
- 13 THE INTERSECTION BETWEEN THE SHANK AND THREAD OF HI-LITE TYPE BOLTS SHALL BE COLD WORKED SUBSEQUENT TO HEAT TREATMENT. THIS PROCESS IS NOT REQUIRED FOR VERI-LITE (VL18 AND VL331), AERO-LITE (AL331), OR ULTRA-LITE (WC331) TYPE BOLTS.
- 14 THE "JT" TRANSITION RADIUS MUST FAIR INTO THE SHANK DIAMETER WITHIN DISTANCE "G" WITH NO VISIBLE LINE, AND EXTEND BELOW THE MAXIMUM "TD" DIAMETER, PER [Figure 1](#).
- 15 GRIP LENGTH, TRANSITION DIMENSIONS ("D", "G" AND "JT"), AND THREAD LEAD SHALL BE INSPECTED USING A COMPARATOR AND OVERLAY CHARTS.
- 16 THREADS ADJACENT TO THE CHAMFER AND GRIP MAY BE INCOMPLETE. INCOMPLETE THREADS MAY BE OUT OF TOLERANCE AT MAJOR, PITCH OR MINOR DIAMETER. LENGTH OF INCOMPLETE THREADS SHALL NOT EXCEED 1.5 PITCHES AT GRIP AND 2.0 PITCHES AT LEAD. TRANSITION AREA AND THREAD FORM WILL ALLOW MATING COLLAR TO ACHIEVE MINIMUM GRIP CONDITION WITHOUT INTERFERENCE.
- 17 THE HEAD ANGLE SHALL BE 100 ± 1 DEGREES WHEN INSPECTED USING AN OPTICAL COMPARATOR.
- 18 RUNOUT OF THE "D" DIAMETER TO THE THREAD MAJOR DIAMETER SHALL BE WITHIN "Y" FIM WHEN HELD ON THE MAJOR DIAMETER OF THE COMPLETE THREADS AND CHECKED ON THE SHANK WITHIN ONE DIAMETER OF THE THREAD RUNOUT.
- 19 PAINT ADHESION TESTS PER [BMS10-85](#) SHALL BE CONDUCTED PERIODICALLY (AT LEAST ONCE IN A SIX MONTH PERIOD). IF THE TEST IS NOT SUCCESSFUL, CORRECTIVE ACTION TO THE PROCESS MUST BE TAKEN PRIOR TO THE COATING OF ADDITIONAL MANUFACTURING LOTS.
- 20 PARTS RELEASED FOR PRODUCTION (AT THE SUPPLIER'S FACILITY) AFTER JUNE 1, 1999 SHALL MEET THE FOLLOWING REQUIREMENT FOR TRANSITION RUN OUT LIMITS:
- FINGER TIGHTEN A FACED OFF GO RING GAGE ON THE THREADS AND MEASURE DIMENSION "M". DIMENSION "M" MUST BE LESS THAN NOMINAL GRIP LENGTH + "L" MAXIMUM (SEE [Figure 1](#) AND [Table I](#)). ACCEPTANCE SAMPLING PER "ATTRIBUTE SAMPLING FOR FINISH INSPECTION AND COMPARATOR MEASUREMENTS" OF [BPS-F-67](#).
- 21 THE SURFACE OF THE RING GAGE SHALL BE GROUND SUCH THAT THE LEAD-IN CHAMFER IS REMOVED (SHARP THREAD START), CONFORMING TO ST1416-13 DRAWING.
- 22 PARTS RELEASED FOR PRODUCTION (AT THE SUPPLIER'S FACILITY) AFTER JUNE 1, 1999 SHALL HAVE THESE PART NUMBERS. PARTS PREVIOUSLY MADE AS VL18 MAY BE USED UNTIL STOCK IS DEPLETED.
- 23 ALCOA FASTENING SYSTEMS (TEMPLE) (FORMERLY FAIRCHILD - TEMPLE) IS NO LONGER IN OPERATION. PARTS MANUFACTURED AT THAT FACILITY UNDER EITHER NAME (CAGE CODE 06950) MAY BE PROCURED AND USED BY BOEING AND ITS SUBCONTRACTORS UNTIL DEPLETED.

- [24] FAIRCHILD FASTENERS (UNRUH) IS NOW ALCOA FASTENING SYSTEMS - INDUSTRY (UNRUH). THIS CHANGE WAS A NAME CHANGE ONLY. STOCK MANUFACTURED UNDER THE FAIRCHILD NAME MAY BE PROCURED AND USED UNTIL DEPLETED.
- [25] HUCK INTERNATIONAL INC (CARSON) IS NOW ALCOA FASTENING SYSTEMS - CARSON. THIS CHANGE WAS A NAME CHANGE ONLY. STOCK MANUFACTURED UNDER THE HUCK NAME MAY BE PROCURED AND USED UNTIL DEPLETED.
- [26] WHERE REQUIRED BY CODE, PAINT ON TOP OF DOMED HEAD ONLY (NON-BEARING SURFACE) INCLUDING THE CENTER DIMPLE, WITH OVERSPRAY ALLOWED ON THE LAND PORTION OF THE HEAD (DEFINED BY DIMENSION "E"). NO OVERSPRAY PERMITTED ON OTHER SURFACES. ALL FASTENER DIMENSIONS APPLY PRIOR TO THE APPLICATION OF PAINT FINISHES IN ACCORDANCE WITH [BAC5684](#).
- [27] FASTENER DIMENSIONS SHOWN APPLY AFTER ALL FINISHES, EXCEPT PAINTS APPLIED IN ACCORDANCE WITH [BAC5684](#).
- [28] FOR PARTS WITH "W" CODE THE CETYL ALCOHOL LUBRICANT MUST BE APPLIED AFTER THE PAINT HAS BEEN APPLIED TO THE BOLT HEAD AND HAS BEEN "PARTIALLY CURED" AS DEFINED IN [BAC5684](#).
- [29] HI-SHEAR CORPORATION IS NOW LISI AEROSPACE - HI SHEAR PRODUCTS. THIS WAS A NAME CHANGE ONLY. PARTS MANUFACTURED UNDER THE HI-SHEAR NAME MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- [30] SMOOTH TRANSITION TO "J" DIAMETER.

PROCUREMENT SPECIFICATION

[BPS-F-67](#), EXCEPT AS NOTED.

INSTALLATION FORCE PER [BMS10-85](#) SHALL BE INSPECTED ON 5 PIECES PER LOT, AFTER LUBRICATION.

PROCESS CAPABILITY
PER [BPS-P-170](#).

KEY CHARACTERISTICS ARE AS FOLLOWS. ALL KEY CHARACTERISTICS APPLY AFTER COATING (BEFORE LUBRICATION OR AFTER REMOVAL OF LUBRICATION).
DIMENSION A' (HEAD OD).

DIMENSION D (SHANK DIAMETER).

DIMENSION H (GAGE PROTRUSION).

THREAD PITCH DIAMETER.

THE MANUFACTURER SHALL IDENTIFY KEY PROCESS PARAMETERS THAT HAVE THE GREATEST EFFECT ON THE KEY CHARACTERISTICS AND DOCUMENT THE NECESSARY PROCESS CONTROLS IN A PROCESS CONTROL DOCUMENT (PCD). EACH KEY CHARACTERISTIC SHALL BE CONTROLLED PER [BPS-P-170](#).

MATERIAL

6Al-4V TITANIUM ALLOY PER [AMS4928](#) OR [AMS4967](#). MINIMUM SHEAR STRENGTH 95 KSI.

FINISH

- LETTER "K" - ALUMINUM PIGMENTED COATING PER [BMS10-85](#), TYPE I, CLASS A. [19](#)
- LETTER "W" - SURFACES INDICATED BY [26](#) (TOP OF BOLT DOMED HEAD) SHALL BE PAINTED IN ACCORDANCE WITH [BAC5684](#), TYPE 1. MAY ONLY BE USED WITH "K" CODED PARTS. (PARTS WHICH HAVE BEEN ALUMINUM PIGMENT COATED PRIOR TO THE APPLICATION OF THE PAINT.)

SURFACE TEXTURE

PER [ASME B46.1](#) PRIOR TO FINISH. CONICAL SURFACE OF HEAD, HEAD TO SHANK FILLET RADIUS, SHANK, TRANSITION AREA, THREAD ROOT AND THREAD FLANKS, 32 MICROINCHES Ra; OTHER SURFACES, 125 MICROINCHES Ra.

LUBRICATION

CETYL ALCOHOL PER [AS87132](#), TYPE I OR TYPE III, GRADE OPTIONAL. [28](#)

MARKING

MARK BOLT HEAD WITH MANUFACTURER'S BASIC NUMBER PLUS DASH NUMBER (DESIGNATES DIAMETER) AND SYMBOL/INSIGNIA PER [Table II](#), PER [MIL-HDBK-57](#), OR REGISTERED WITH THE U.S. PATENT AND TRADEMARK OFFICE (PTO) OF THE U.S. DEPARTMENT OF COMMERCE. MARKING INDENTED .006 MAXIMUM, ARRANGEMENT OPTIONAL.

IDENTIFY WITH ORANGE DYE ON THREAD END BEFORE LUBRICATION. COVERAGE OF DYE MAY INCLUDE THE CHAMFER DIMENSION PLUS ONE PITCH.

PACKAGING

IN ADDITION TO THE PACKAGING REQUIREMENTS OF [BPS-F-67](#), ALL PARTS WITH PAINTED HEADS SHALL BE PACKAGED TO PREVENT NICKING OR SCRATCHING OF PAINTED SURFACES OR OTHER DAMAGE DURING HANDLING, TRANSPORTATION, OR STORAGE.

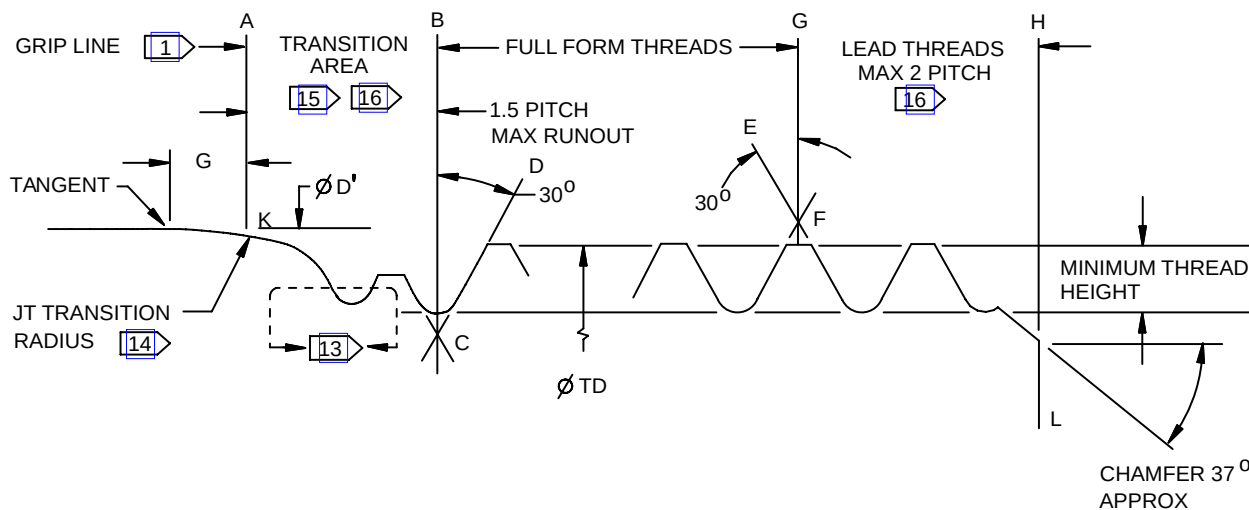
INSPECTION REQUIREMENTS

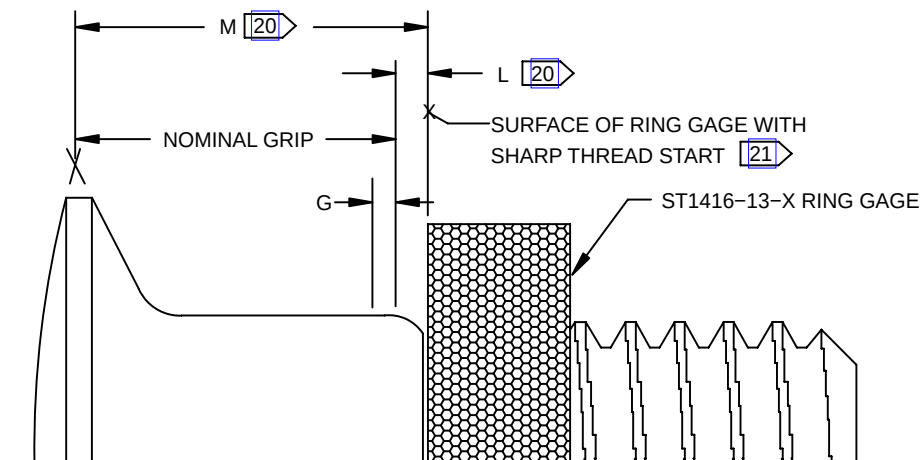
FIGURE 1 - DETERMINATION OF INCOMPLETE THREADS BY COMPARATOR

CHART REQUIREMENTS (Figure 1)

1. FOR RUNOUT THREADS: HORIZONTAL PARALLEL LINES TO DEFINE MINIMUM THREAD HEIGHT, WHICH IS ONE-HALF THE DIFFERENCE BETWEEN MINIMUM MAJOR DIAMETER AND MAXIMUM MINOR DIAMETER; VERTICAL LINE "AK" LOCATED TO LEFT OF VERTICAL LINE "BC"; VERTICAL LINE "BC" INTERSECTING 30 DEGREE FLANK ANGLE LINE "CD" AT POINT "C".
2. FOR LEAD THREADS: HORIZONTAL LINES AS DESCRIBED ABOVE; VERTICAL LINE "GF" INTERSECTION 30 DEGREES FLANK ANGLE LINE "EF" AT POINT "F"; VERTICAL LINE "HL" LOCATED 2 PITCH DISTANCE TO RIGHT OF "GF".

PROCEDURES (Figure 1)

1. FOR RUNOUT THREADS: ROTATE FASTENER TO FIND FIRST COMPLETE THREAD NEAREST TO GRIP. AT THIS POINT, THREAD HEIGHT EQUALS MINIMUM HEIGHT AS INDICATED ON CHART. THEN MOVE SHADOW LONGITUDINALLY UNTIL LEFT FLANK OF FIRST COMPLETE THREAD PROFILE COINCIDES WITH LINE "CD". RUNOUT THREADS SHALL THEN FALL BETWEEN LINE "AK" AND "BC" TO BE ACCEPTABLE.
2. FOR LEAD THREADS: ROTATE FASTENER TO FIND FIRST COMPLETE THREAD NEAREST TO END. AT THIS POINT COMPLETE THREAD HEIGHT EQUALS MINIMUM HEIGHT AS INDICATED ON CHART. THEN MOVE SHADOW LONGITUDINALLY UNTIL RIGHT FLANK OF FIRST COMPLETE THREAD PROFILE COINCIDES WITH LINE "EF". LEAD THREADS SHALL THEN FALL BETWEEN LINE "GF" AND "HL" TO BE ACCEPTABLE.

**FIGURE 2 - TRANSITION RUN OUT LIMITS****PROCUREMENT**

AIR INDUSTRIES CORP (CAGE CODE 06725)

ALCOA FASTENING SYSTEMS - CARSON OPERATIONS (CAGE CODE 17446) [25]

ALCOA FASTENING SYSTEMS - INDUSTRY (TEMPLE) (CAGE CODE 06950) [23]

ALCOA FASTENING SYSTEMS - INDUSTRY (UNRUH) (CAGE CODE 1RC86) [24]

LISI AEROSPACE - HI-SHEAR PRODUCTS (CAGE CODE 73197) [29]

WEST COAST AEROSPACE (CAGE CODE 60516)

FASTENERS WITH HEADS PAINTED IN ACCORDANCE WITH [BAC5684](#), SHALL HAVE THE PAINTS APPLIED BY ONLY APPROVED SOURCES LISTED IN THE [BAC5684](#) QPL.

THE MANUFACTURERS LISTED IN [BPS-F-67SUP](#) AND THEIR AUTHORIZED DISTRIBUTORS ARE THE ONLY APPROVED SOURCES FOR THE ABOVE QUALIFIED PRODUCTS. SEE [BPS-F-67SUP](#) FOR PLANT ADDRESSES. NO CHANGES IN PRODUCT DESIGN, BASIC METHODS OF MANUFACTURE, PLANT SITE OR QUALITY LEVEL SHALL BE MADE WITHOUT PRIOR NOTIFICATION AND PRIOR APPROVAL IN WRITING FROM THE BOEING COMPANY. MANUFACTURERS OF COMPETITIVE PRODUCTS MAY APPLY TO A SUPPLIER MANAGEMENT DEPARTMENT OF THE BOEING COMPANY FOR QUALIFICATION. IF A MANUFACTURER IS SHOWN ON THIS STANDARD, BUT NOT LISTED IN THE SUPPLEMENT, CONTACT THE DIVISIONAL ENGINEERING STANDARDS FOCAL POINT OR ENGINEERING STANDARDS FOR VERIFICATION.

THIS IS A MANUFACTURER-DESIGNED PRODUCT. BOEING MAKES NO REPRESENTATION WHATEVER REGARDING PATENT OR OTHER RIGHTS AFFECTING THE PRODUCT. THE LISTING OF ANY SUPPLIER DOES NOT IMPLY ANY DETERMINATION BY THE BOEING COMPANY OR BY ANY OTHER LISTED MANUFACTURER AS TO THE RIGHTS OF SUCH MANUFACTURER.

THE MANUFACTURER, ALCOA FASTENING SYSTEMS, STATES THAT PATENT NO. 4,735,537 IS PERTINENT TO ITS PRODUCT.

USAGE AND APPLICATION INFORMATION

THESE BOLTS AND THE MATING COLLARS WITH COUNTERBORE ARE DESIGNED SO THAT NOMINAL GRIP DIMENSIONS EQUAL THE MAXIMUM MATERIAL THICKNESS (OR MAXIMUM GRIP RANGE). WHEN MATERIAL THICKNESS EQUALS MINIMUM GRIP, A NOMINAL UNTHREADED SHANK LENGTH OF .062 ENTERS THE COUNTERBORE OF THE COLLAR.

BACB30YP BOLTS HAVE A HEAD CONFIGURATION IDENTICAL TO THAT OF [BACB30XT](#) AND [BACB30XU](#) LOCKBOLTS. THESE CLOSE TOLERANCE BOLTS ARE FOR APPLICATIONS WHERE CROWN HEIGHT CONTROL AND FLUSHNESS ARE BOTH REQUIRED. DESIGNERS MAY ALSO USE BACB30YP BOLTS FOR THE SAME TYPE OF APPLICATIONS AS [BACB30VU](#).

FOR REPAIR, SEE [BACB30NW](#) "X" AND "Y" OVERSIZE FOR USAGE.

SEE [BACD2074](#) FOR FASTENER CODES.

INSTALL PER [BAC5004-2](#) OR [BAC5063-2](#) AS APPLICABLE.

BACB30YP FASTENERS WHERE THE HEAD HAS BEEN PRE-PAINTED IN ACCORDANCE WITH [BAC5684](#) IS INTENDED TO BE USED ONLY IN APPLICATIONS WHERE THE STRUCTURE HAS BEEN PRE-PAINTED PRIOR TO THE INSTALLATION OF THE FASTENER.

DO NOT USE THE BACB30YP WHERE THE HEAD HAS BEEN PRE-PAINTED IN ACCORDANCE WITH [BAC5684](#), IN APPLICATIONS WHERE IT WILL BE INSTALLED IN A INTERFERENCE FIT HOLE REQUIRING THE FASTENER TO BE DRIVEN INTO THE HOLE.

CODING

FIRST DASH NUMBER DESIGNATES NOMINAL THREAD SIZE PER [Table I](#).

LETTER "K" FOLLOWING THE FIRST DASH NUMBER DESIGNATES ALUMINUM PIGMENTED COATING WITH CETYL ALCOHOL LUBRICATION.

SECOND DASH NUMBER DESIGNATES NOMINAL GRIP LENGTH IN .0625 INCREMENTS. [1](#)➤

LETTER "W" FOLLOWING THE SECOND DASH NUMBER DESIGNATES TOP OF HEAD PAINTED IN ACCORDANCE WITH [BAC5684](#) TYPE 1. [26](#)➤ MAY BE USED WITH ALUMINUM PIGMENT COATED BOLTS ONLY. ("K" CODES.)

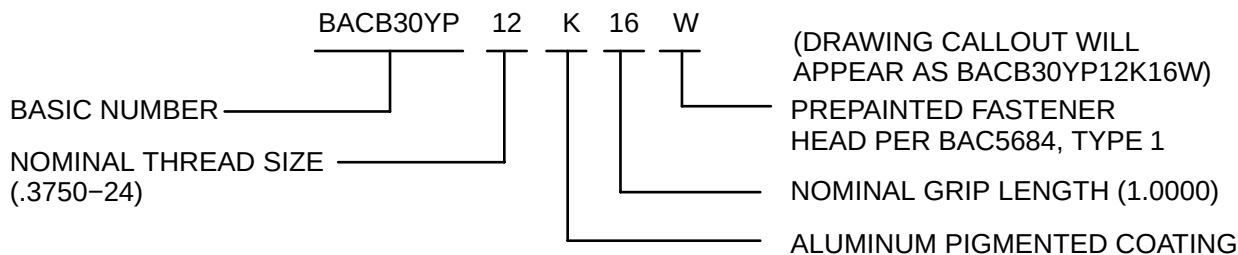
EXAMPLE OF PART NUMBER

TABLE II

MANUFACTURER	MANUFACTURER'S BASIC PART NUMBER	MANUFACTURER'S SYMBOL/INSIGNIA
AIR INDUSTRIES	HST331	⌘
ALCOA FASTENING SYSTEMS - INDUSTRY	VL331	O OR VS
LISI AEROSPACE - HI SHEAR PRODUCTS	HST331	H, HS, OR NONE
ALCOA FASTENING SYSTEMS - CARSON	AL331 	⌘
WEST COAST AEROSPACE	WC331	WC
MANUFACTURER'S BASIC NUMBER AND SYMBOL/INSIGNIA LISTED ARE FOR HEAD MARKING AND ARE NOT TO BE USED FOR DRAWING CALLOUT.		

D-590 NOTES

SEE [D-590-PREFACE](#) FOR INACTIVATION DEFINITIONS. SEE [00000-D-590-INFO](#) FOR LINKS TO SUPERSESSION CLASS DEFINITIONS AND FAQ.